

# Vira Rithy



rithyvira021@gmail.com



+855 69-625-363 | 95-950-349



Portfolio



@rithyvira



Phnom Penh, Cambodia



## PROFILE

Passionate about Data Science, AI, GenAI, and Machine Learning, with expertise in machine learning, data analytics, and cutting-edge AI applications. Skilled in extracting insights from complex datasets and delivering impactful solutions. Driven by innovation, collaboration.



## EDUCATION

### DataU Academy

Data Specialist

08/2023 - 01/2024

### Institute of Technology of Cambodia

Bachelor of Engineering (BEng), Data Science

02/2021 - 07/2025



## CERTIFIED LICENSE

SDOE : DevOps Engineering

SalaCyber

SHOW CREDENTIAL

IBM AI Engineering Specialization

IBM

SHOW CREDENTIAL

AI Engineering Bootcamp

Maven - AI Makerspace

SHOW CREDENTIAL

Machine Learning with Python-From Linear Models to Deep Learning

MITx MicroMasters

SHOW CREDENTIAL



## PERSONAL PROJECTS

### • Multi-Agent RAG Marketing Content Enhancer"

The Multi-Agent RAG platform drafts, edits, cites, and adapts, ensuring accurate content with future visual and scalable features.

OpenAI

LANGCHAIN

LANGGRAPH

SHOW PROJECT



## PROFESSIONAL EXPERIENCE

### ZTOA Group

Freelance, Full-Stack AI Developer

04/2025 - 06/2025

- **Email Application System Dashboard"** **NODE.JS** **EXPRESS** **Javascript** **POSTGRES** **SHOW PROJECT**  
Built a full-stack AI-powered email dashboard that imports contact emails, generates personalized content from domain websites, and automates campaign delivery. Uses fallback templates for generic emails.Tech stack: React, Node.js, Express, PostgreSQL, Apache
- **Automated Scrape Email and Generated SEO Marketing Report"** **n8n** **OpenAI**  
Built an end-to-end n8n workflow to scrape business websites from Google Maps, extract emails, generate structured SEO reports, and auto-send personalized result, using JavaScript, PostgreSQL, OpenAI, and SMTP.

### DataTicon

AI Engineer

02/2024 - 06/2025

- **Khmer Speech Recognition "Speech-to-Text EN-KH"** **PYTORCH** **HUGGING FACE** **SHOW PROJECT**  
Fine-tuned OpenAI Whisper-Small for KH and EN, achieving a 37% WER. Deployed on Hugging Face with a Gradio-integrated UI.
- **AI Data Analysis "Text-to-SQL, Voice-to-Chart"** **LLM** **CHAINLIT** **PYTHON** **SHOW PROJECT**  
Built an AI tool using dynamic databases for text-to-SQL and text-to-chart outputs, enabling quick insights and efficient data visualization
- **Machine Learning "Forecasting Cambodia Tourism"** **ENSEMBLE** **PYTHON** **MLFLOW** **SHOW PROJECT**  
Achieved 97% accuracy in forecasting monthly visitors (e.g., August 2024: Predicted 548K vs. Actual 547K) using an ARIMA-XGBoost ensemble with MLflow tracking.

### DataU Academy

Data Specialist

08/2023 - 01/2024

- **Data Warehousing: "Global Cyclistic Company"** **SQL** **ETL** **SHOW PROJECT**  
Transform massive datasets from diverse sources for accessible insights and powerful sales optimization analytics.
- **AI-Powered Sales Analysis & Forecasting** **PYTHON** **STREAMLIT** **SHOW PROJECT**  
Developed a sales forecasting solution for Olist using SARIMAX, XGBoost, and LSTM (top performer), integrating Streamlit and Power BI for revenue, stock, and marketing insights.
- **Hazard Vulnerability Dashboard** **POWER BI** **SHOW PROJECT**  
Analyzed Cambodia's hazard vulnerability index (2014-2019) at provincial/commune levels, assessing risks from storms, floods, and droughts, with Khmer and English support.



## SKILLS

### TECHNICAL

#### Programming

Python for Data processing, ML model development, API integration, and PyTorch for deep learning.

#### DevOps

Experienced in CI/CD, Docker, Kubernetes, and ML pipeline automation using

#### Power BI

Data visualization, reporting, and ETL tools for data integration & analysis.

#### Math

Statistics, Probability, Linear Algebra, Probabilistic Graphical Models, Calculus, Numerical Analysis, Discrete Maths

#### LLMs

Hands-on experience with Prompt Engineering, RAG, Fine-Tuning and Multi-Agent Systems.

### INTERPERSONAL

#### LEADERSHIP

Inspiring others through positive example, effective communication, and collaborative decision-making.

#### COMMUNICATION

Inspiring others through positive example, effective communication, and collaborative decision-making.

#### PROBLEM SOLVING

Prioritize analytical thinking, creativity, and a collaborative mindset to find innovative and effective solutions.

#### BUSINESS CONCEPT

Centers on delivering innovative products/services that address market needs, backed by a commitment to quality, sustainability, and exceptional customer experiences.